MAB and UCB

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- N arms. Arm i yields an average reward r_i
- But when you pull arm i for the ith time, the reward is a r.v. X_{it} with $E(X_{it}) = r_i$, $X_{it} \in [0, 1]$
- Let $n_i(t)$ be the # times arm i has been pulled upto and including time t.

- R_T = expected regret after T pulls

$$= \sum_i \Delta_i E(n_i(T)),$$

where $\Delta_i = r_{max} - r_i$, where $r_{max} = \max_j r_j$.

Explore and Commit Algorithm

Let N=2. Pull each arm m times, compute the empirical rewards

$$\hat{r}_i = \frac{\sum_{t=1}^{m} X_{it}}{m}$$

- Pull arm 1 if $\hat{r}_1 > \hat{r}_2$

- Pull arm 1 if $\hat{r}_1 > \hat{r}_2$
- Else pull arm 2.

- For the purpose of analysis assume $r_1 > r_2$

$$R_T = \underbrace{\Delta_2 m}_{\text{due to pulling of the wrong arm } m \text{ times}} + (T-m)\, P(\hat{r}_2 \geq \hat{r}_1)$$

$P(\hat{r}_2 \geq \hat{r}_1) = \varepsilon_m$ (may not be easily computable but such an $\varepsilon_m > 0$ exists.)

$$\Rightarrow R_T = \Delta_2 m + (T-m)\varepsilon_m = O(T)$$

Can we reduce regret? How about choosing m as a function of T?

$$R_T \leq \Delta_2 m + T\, P(\hat{r}_2 \geq \hat{r}_1)$$

$$= \Delta_2 m + T\, P\left(\sum_{t=1}^{m} \frac{X_{2t} - X_{1t}}{m} \geq 0 \right)$$

Hoeffding bound: If $Y_i \in [a_i, b_i]$, independent, then

$$P\left(\frac{1}{m}\sum_{i=1}^{m}\left(X_i - E(Y_i)\right) \geq \epsilon\right) \leq \exp\left(\frac{-2m^2\epsilon^2}{\sum_{i=1}^{m}(b_i - a_i)^2}\right)$$

$$\Rightarrow \quad R_T \leq \Delta_2 m + T\, P\left(\frac{\sum_{t=1}^{m}(X_{2t} - X_{1t}) + \Delta_2}{m} \geq \Delta_2\right)$$

$$\leq \Delta_2 m + T \exp\left(-2m\Delta_2^2\right)$$

To find the optimal m:

$$\Delta_2 + T(-2\Delta_2^2)\, e^{-2m\Delta_2^2} = 0$$

$$e^{-2m\Delta_2^2} = \frac{1}{2T\Delta_2}$$

$$m = \frac{1}{2\Delta_2^2} \ln 2T\Delta_2$$

$$\Rightarrow R_T \leq \frac{1}{2\Delta_2} \ln(2T\Delta_2^2) + \frac{1}{2\Delta_2}$$

$$= O\left(\frac{\ln T}{\Delta_2}\right)$$

Disadvantages of the algorithm: To choose n, we need to know T and a lower bound on Δ_2. UCB overcomes these issues.

UCB:

- Pull each arm once. (initialization)
- For $t = N, N+1, \ldots$
  - Let $n_i(t)$ be the # times arm i has been pulled in time slots up to t
  - Define $UCB_i(t) = \hat{r}_i(t) + \sqrt{\frac{2 \ln t}{n_i(t)}}$,

    where $\hat{r}_i(t) = \frac{\sum_{\tau=0}^{t} X_{i\tau}}{n_i(t)}$
  - Pull the arm with the largest $UCB_i(t)$ at time $t+1$

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Analysis: for the analysis w.l.o.g. assume $i^* = 1$

Suppose $\hat{r}_i \le r_i + \sqrt{\frac{2\ln t}{n_i(t)}}$ for $i \neq 1$

and $\hat{r}_1 \ge r_1 - \sqrt{\frac{2\ln t}{n_i(t)}}$

$\Rightarrow$

$$\hat{r}_i \le r_i + \sqrt{\frac{2\ln t}{n_i(t)}} \quad (A)$$

and

$$\hat{r}_1 \ge r_1 - \sqrt{\frac{2\ln t}{n_i(t)}} \quad (B)$$

In this case, $UCB_i(t) \le r_i + 2\sqrt{\frac{2\ln t}{n_i(t)}}$

and

$$UCB_1(t) \ge r_1$$

$\Rightarrow$ i cannot be pulled if

$$\Rightarrow \quad \Delta_i > 2\sqrt{\frac{2\ln T}{n_i(t)}} \quad \Rightarrow \quad n_i(t) > \frac{8\ln T}{\Delta_i^2}$$

- Let I_{t+1} be the arm pulled at time $(t+1)$

$$\Rightarrow \left\{ I_{t+1} = i, \quad n_i(t) > \frac{8\ln T}{\Delta_i^2} \right\}$$

$$\Rightarrow \quad A^c \cup B^c$$

$$\Rightarrow \quad P(I_{t+1}=i) = P\left(I_{t+1}=i, \; n_i(t) \le \frac{4\ln T}{\Delta_i}\right) + P(A_t^c \cup B_t^c)$$

$$\Rightarrow \quad E(n_i(T)) = 1 + \sum_{t=N+1}^{T} P\left(I_t = i, \; n_i(t-1) \le \frac{4\ln T}{\Delta_i}\right) + \sum_{t=N+1}^{T} \left(P(A_{t-1}^c) + P(B_{t-1}^c)\right)$$

$$\le 1 + \frac{8\ln T}{\Delta_i^2} + \sum_{t=1}^{T} \left(P(A_{t-1}^c) + P(B_{t-1}^c)\right)$$

since $I_t = i$ can only be true of $n_i(T-1)$ times

times

$$P(A_{t-1}^c) = P\left(\sum_{\tau=1}^{n_i(t-1)} \frac{1}{n_i(t-1)} (X_{i\tau} - \mu_i) > \sqrt{\frac{2 \ln t}{n_i(t-1)}} \right)$$

$$\leq \sum_{\ell=1}^{t-1} P\left(\frac{1}{\ell} \sum_{\tau=1}^{t-1} (X_{i\tau} - \mu_i) \geq \sqrt{\frac{2 \ln t}{\ell}} \right)$$

$$\leq \sum_{\ell=1}^{t-1} \exp\left(-2 \left(\frac{2 \ln t}{\ell} \right) \ell \right)$$

$$= \frac{(t-1)}{t^4} \leq \frac{1}{t^3}$$

Similarly for $P(B_{t-1}^c)$

$$\Rightarrow E(n_i(T)) \leq 1 + \frac{8 \ln T}{\Delta_i^2} + \sum_{t=1}^{T} \frac{2}{t^2} \quad \left(\frac{1}{t^2} \geq \frac{1}{t^3} \ \forall t \geq 1 \right)$$

$$\leq 1 + \frac{8 \ln T}{2} + \frac{\pi^2}{3}$$

$$\leq 1 + \frac{8 \ln T}{\Delta_i^2} + \frac{\pi^2}{3}$$

$$\Rightarrow \quad \boxed{R(T) \leq \sum_{i=2}^{N} \left(\frac{8 \ln T}{\Delta_i} + \Delta_i \left(1 + \frac{\pi^2}{3} \right) \right)}$$